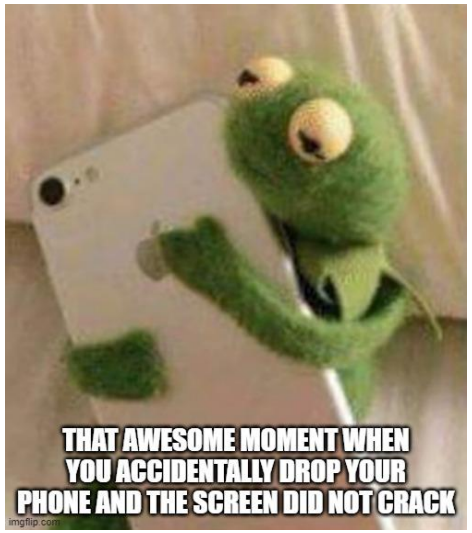


CARBON FIBER/MODULAR BAMBOO COMP PHONE CASE

Logiit Mugunthan

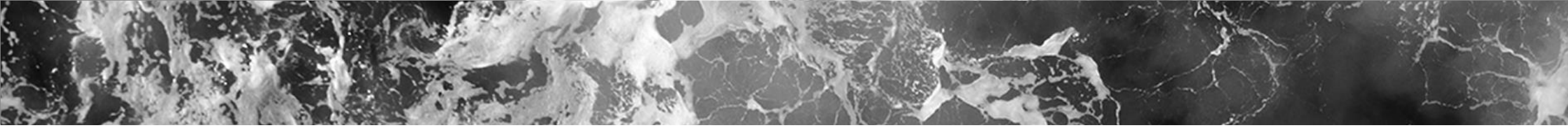
THE PROBLEM



According to a *New York Post* article, Americans drop their phones 4 times a week on average.

Most phone cases don't survive very long.

Screen protectors are not long lasting.



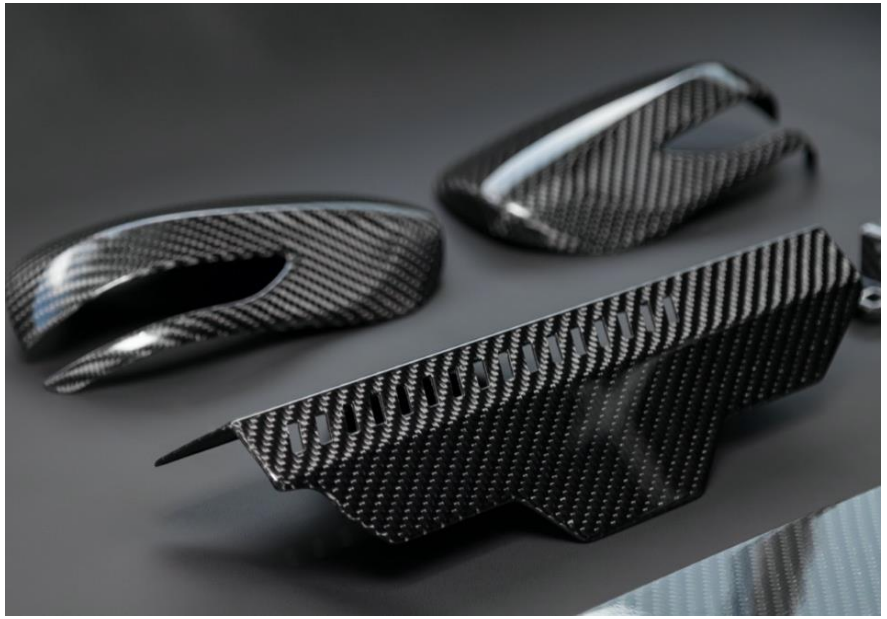
THE SOLUTION

A sturdier phone case

Most phone cases nowadays are made from plastic, silicon, leather (very rarely), or metal. Leather creases easily and is often replaced within a year or two and it is not particularly durable to hard falls. Metal ones are durable; however, they are often very bulky and heavy. Plastic is easily breakable and likewise, silicon can easily tear and will not withstand most heavy falls.

A more durable screen protector

Tradition screen protectors are made from plastic, or tempered glass. A typical tempered glass screen protector lasts about 6 – 12 months, or 1-2 years with *extreme* care. Plastic lasts less, about 3-6 months, or 6-12 months with extreme care.



Stiff

Carbon fiber is extremely deformation resistant.

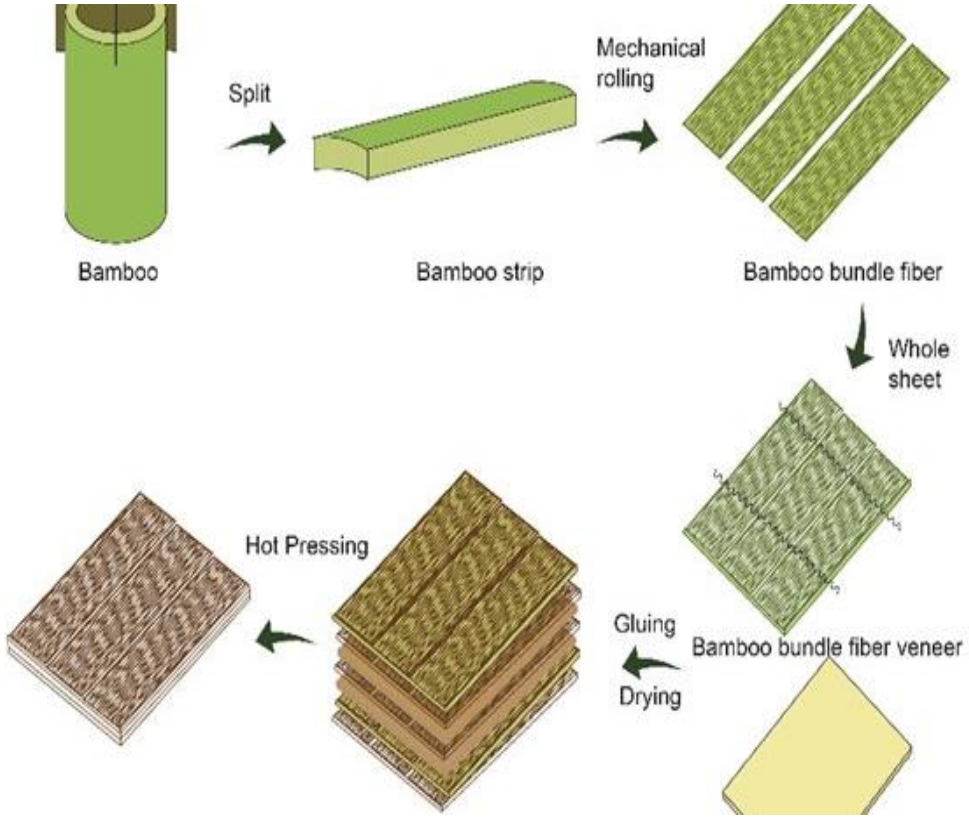
Design flexibility

Carbon fiber can be molded into complex shapes, such as that of a phone case.

Low impact resistance

Carbon fiber can't dissipate energy as effectively so it will not absorb the impact generated by a fall.

CARBON FIBER



MODULAR BAMBOO

Shock absorbent

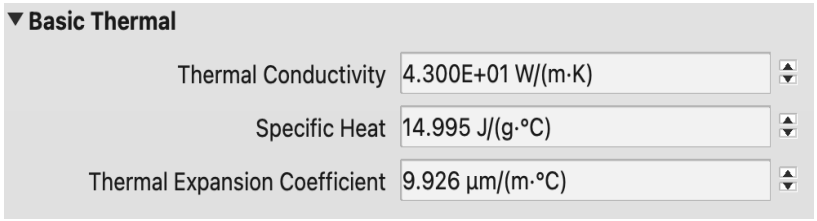
Modular bamboo is highly shock absorbent due to its natural flexibility.

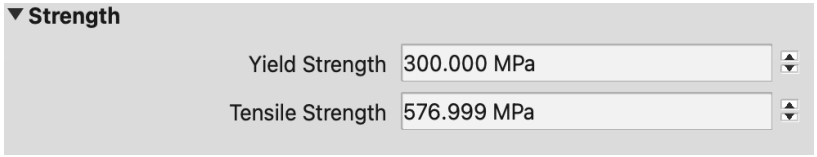
Light weight & strong

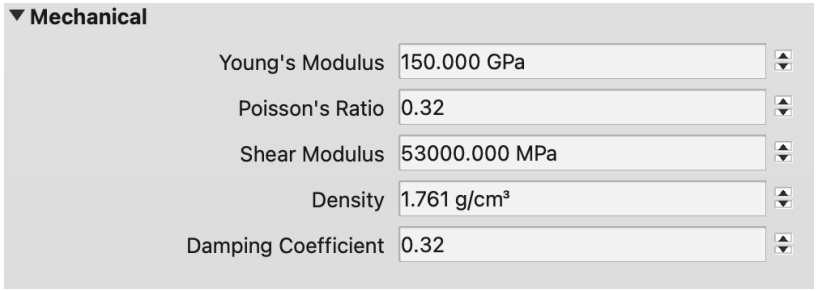
Modular bamboo is lightweight, due to its natural hollow structure, and it is strong due to its processing.

Easy to assemble

Modular bamboo can readily be created and integrated into anything easily.







CARBON MB COMP

Shock absorbing and distributing

The damping coefficient is high for this comp material which means that it will effectively absorb and dissipate impact shocks such as those from dropping your phone on the ground.

Strength

The yield strength and tensile strength are high enough that it can withstand drops and heavy impact without any structural damage. Combined with its high damping coefficient makes it extremely safe for your phone.

Rigidness

This material is rigid so it might be difficult to put the phone in and get it out, but if tempered correctly, it can be made more flexible. Its rigidness also helps maintain structural integrity, making sure that your phone is kept nice and safe.



TRANSPARENT ALUMINUM

Heat dissipation

This material is very good at dissipating heat that can be produced from the screen.

Downsides

This material is brittle and can be scratched which is why this might be a good material as part of a multi-layered screen protector. This material can be the inner layer, focusing more on heat dissipation, while there is a stronger, more resilient outer layer.

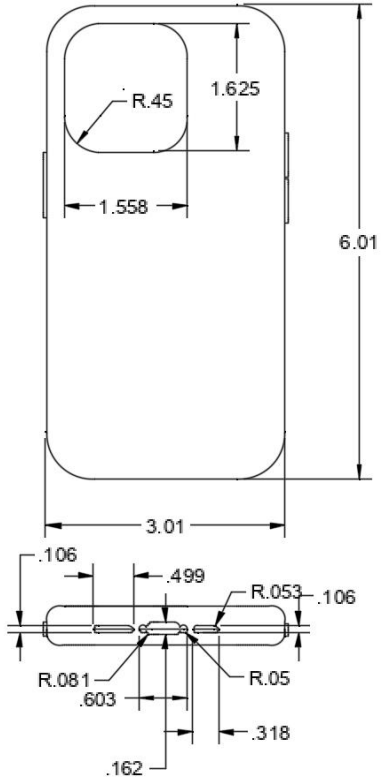
The Product



PARTS LIST				
ITEM	QTY	PART NUMBER	MASS	MATERIAL
1	1	CARBON MB COMP PHONE CASE - MOCKUP V2	2.183 ouncemass	CARBON MB COMP
2	1	TRANSPARENT ALUMINUM SCREEN PROTECTOR V1	1.82 ouncemass	ALUMINUM OXYNITRIDE

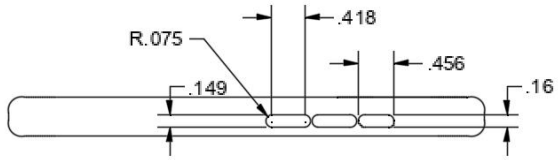


SKETCHES AND PRODUCT DRAWINGS



Tolerances
X.X - ± .2
X.XX - ± .02
X.XXX - ± .002





MANUFACTURING & POTENTIAL ISSUES

CARBON FIBER

- high cost of raw materials: resin, carbon fibers
- labor intensive
- curing time: several hours, slows production
- export restrictions due it being primarily used in aerospace/defense industries
- polyacrylonitrile (PAN) is part of the fiber that makes up carbon fiber and it is derived from oil, making its cost dependent on oil costs and it is mainly produced in the US and Japan
- resin is mainly produced by South Korea, China, US, and Japan
- Producing my own carbon fiber would be better due to import restrictions and availability of resources inside the US.

MODULAR BAMBOO

- moisture sensitive
- specialized processing equipment required
- only produced in areas with optimal climate (Southeast Asia)
- transportation costs
- creating a farm/harvesting can negatively affect biodiversity/compete with food crops
- I would outsource this material even though the transportation costs may be high because it is not practical to grow and produce bamboo in the US

TRANSPARENT ALUMINUM

- high cost of production
- specialized machines/tools required to grind it
- long production time
- few suppliers
- technical expertise required
- processing infrastructure is expensive
- I would outsource this material because I think that the cost to set up a facility and specialized machines as well as hiring specialized engineers and workers is way more than simply importing them from one of the few suppliers in the world.

THANK YOU!